CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 6: Simultaneous Linear Equations

Topic-wise short notes with solved examples

Solving a pair of linear equations in two variables by substitution, elimination and cross-multiplication, equations that reduce to linear form, and word problems.

1 A pair of linear equations

- A **linear equation in two variables** has the form $ax + by + c = 0$, where a and b are not both zero.
- Two such equations taken together are **simultaneous linear equations**. A **solution** is a pair (x, y) that satisfies both.
- Always check the answer by substituting in **both** equations.

2 Method of substitution

- From one equation, express one variable in terms of the other.
- Substitute this in the other equation and solve the resulting equation in one variable.
- Substitute back to find the second variable.

Example 2.1 Solve $x + y = 7$ and $3x - 2y = 11$ by substitution.

SOLUTION

From the first, $y = 7 - x$.

$3x - 2(7 - x) = 11$ gives $5x = 25$, so $x = 5$.

$x = 5, y = 2$

3 Method of elimination

- Multiply the equations by suitable numbers so that the coefficients of one variable become **equal in size**.
- **Add** the equations if those coefficients have opposite signs; **subtract** if they have the same sign.
- Solve for the remaining variable and substitute back.

Example 3.1 Solve $3x + 4y = 10$ and $2x - 3y = 1$ by elimination.

SOLUTION

Multiply the first by 3 and the second by 4: $9x + 12y = 30$ and $8x - 12y = 4$.

Add: $17x = 34$, so $x = 2$. Then $6 + 4y = 10$.

$x = 2, y = 1$

Example 3.2 Solve $41x + 53y = 135$ and $53x + 41y = 147$.

SOLUTION

Add: $94x + 94y = 282$, so $x + y = 3$. Subtract: $12x - 12y = 12$, so $x - y = 1$.

$x = 2, y = 1$

4 Method of cross-multiplication

- For $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$:

$$\frac{x}{b_1c_2 - b_2c_1} = \frac{y}{c_1a_2 - c_2a_1} = \frac{1}{a_1b_2 - a_2b_1}$$

- Write both equations with zero on the right before using the formula.
- The method works when $a_1b_2 - a_2b_1 \neq 0$.

Example 4.1 Solve $2x + 3y - 17 = 0$ and $3x - 2y - 6 = 0$ by cross-multiplication.

SOLUTION

$$\frac{x}{(3)(-6) - (-2)(-17)} = \frac{y}{(-17)(3) - (-6)(2)} = \frac{1}{(2)(-2) - (3)(3)}$$

$$\frac{x}{-52} = \frac{y}{-39} = \frac{1}{-13}$$

$x = 4, y = 3$

5 Equations reducible to linear form

- If the variables appear as $\frac{1}{x}$ and $\frac{1}{y}$, put $\frac{1}{x} = a$ and $\frac{1}{y} = b$, solve for a and b, then invert.
- If they appear as $\frac{1}{x+y}$ and $\frac{1}{x-y}$, use these as the new variables in the same way.
- Equations with xy on the right, such as $2x + 3y = 5xy$: divide every term by xy.

Example 5.1 Solve $\frac{2}{x} + \frac{3}{y} = 13$ and $\frac{5}{x} - \frac{4}{y} = -2$.

SOLUTION

With $a = \frac{1}{x}$, $b = \frac{1}{y}$: $2a + 3b = 13$ and $5a - 4b = -2$.

Multiply by 4 and 3 and add: $23a = 46$, so $a = 2$ and $b = 3$.

$x = \frac{1}{2}, y = \frac{1}{3}$

Example 5.2 Solve $\frac{5}{x+y}-\frac{2}{x-y}=-1$ and $\frac{15}{x+y}+\frac{7}{x-y}=10$.

SOLUTION

With $a=\frac{1}{x+y}$, $b=\frac{1}{x-y}$: $5a-2b=-1$ and $15a+7b=10$.

Three times the first: $15a-6b=-3$. Subtract: $13b=13$, so $b=1$ and $a=\frac{1}{5}$.

$x+y=5$ and $x-y=1$

$x=3, y=2$

6 Word problems

- Choose the two unknowns, form two equations from the conditions, solve and check against the words.

Type	How to set up
Two-digit number	Tens digit x, units digit y: number $10x+y$, reversed $10y+x$
Fraction	Numerator x, denominator y
Ages	Present ages x and y; after t years $x+t, y+t$
Boat and stream	Speed downstream $x+y$, upstream $x-y$
Fixed charge plus rate	Total = fixed charge + rate $\times$ quantity

Example 6.1 The sum of the digits of a two-digit number is 9. The number obtained by reversing the digits is 27 more than the number. Find the number.

SOLUTION

$x+y=9$ and $(10y+x)-(10x+y)=27$, that is $y-x=3$.

$x=3, y=6$; **the number is 36**

Example 6.2 A fraction becomes $\frac{1}{2}$ when 1 is added to the numerator and becomes $\frac{1}{3}$ when 1 is added to the denominator. Find it.

SOLUTION

$2(x+1)=y$ and $3x=y+1$. So $3x=2x+3$.

$x=3, y=8$; **the fraction is** $\frac{3}{8}$

Example 6.3 A boat goes 30 km downstream in 2 hours and 18 km upstream in 2 hours. Find the speed of the boat in still water and the speed of the stream.

SOLUTION

$x + y = 15$ and $x - y = 9$

Boat 12 km/h; stream 3 km/h

Example 6.4 Five years ago a man was 7 times as old as his son. Five years from now he will be 3 times as old. Find their present ages.

SOLUTION

$x - 5 = 7(y - 5)$ and $x + 5 = 3(y + 5)$, that is $x - 7y = -30$ and $x - 3y = 10$.

Subtract: $4y = 40$

Son 10 years, father 40 years

7 Common mistakes

- Changing the sign of only some terms when subtracting one equation from another.
- Multiplying only one side of an equation.
- In cross-multiplication, leaving the constants on the right-hand side.
- Giving a and b as the answer in reducible equations instead of x and y.
- Writing a two-digit number as xy or $x + y$ instead of $10x + y$.
- Not checking the solution in the second equation.

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